

Discrete Mathematics Spring 2026

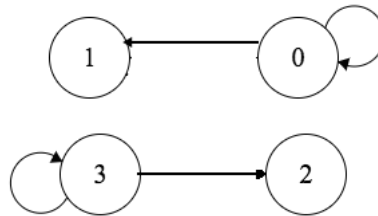
Homework 6: Relations

Due Sunday March 15 @11:59:00pm Eastern

Show all your work to receive full credit

1. Relation properties

Let R be a relation on the set $A = \{0, 1, 2, 3\}$ represented by the graph below.



Check all that apply and explain in one sentence. Explain every property whether you check it or not.

- (a) (5 points) R is Reflexive. ☐ Yes ☐ No

Explanation: _____

- (b) (5 points) R is Irreflexive. ☐ Yes ☐ No

Explanation: _____

- (c) (5 points) R is Symmetric. ☐ Yes ☐ No

Explanation: _____

- (d) (5 points) R is Anti-symmetric. ☐ Yes ☐ No

Explanation: _____

- (e) (5 points) R is Transitive. ☐ Yes ☐ No

Explanation: _____

- (f) (5 points) R is an equivalence relation. ☐ Yes ☐ No

Explanation: _____

2. (25 points) Equivalence relations

Let the set $A = \{3, 4, 5, 6, 7, 8\}$. Let the relation R be defined as follows:

For all $x, y \in A$:

$$x R y \Leftrightarrow 3 \mid (x - y)$$

Prove that R is an equivalence relation. Please do not use an exhaustive proof (enumeration). Show all your work.

3. Congruence relations

Recall the congruence relation of numbers, congruence modulo n .

Let's define R_3 the binary relation of congruence modulo 3 on $\mathbb{Z}$.

$$R_3 = \{(x, y) \mid (x, y) \in \mathbb{Z}^2 \wedge x \equiv y \pmod{3}\}$$

- (a) (9 points) Calculate $[0], [1], [2], [3], [4], [5]$ (You can enumerate the elements).
- (b) (8 points) How many distinct equivalence classes are there? Justify your answer.
- (c) (8 points) Are the equivalence classes disjoint (empty intersection)? What is their union?

4. (25 points) Proof on relations

Let R , S , and T be relations from A to B . Prove using element-chasing that:

$$R \subseteq S \text{ and } R \subseteq T \Rightarrow R \subseteq S \cap T$$